

Vendor Performance Analysis

Exploratory Data Analysis (EDA)

Author

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Introduction

This notebook presents the Exploratory Data Analysis (EDA) of a Vendor Performance dataset.

The primary objective is to understand vendor performance, procurement efficiency, profitability, and inventory behavior before developing an interactive Power BI dashboard.

The analysis focuses on identifying business trends, detecting performance gaps, and generating data-driven insights to support procurement and inventory management decisions.

Objectives

- Understand the structure of the dataset
- Assess data quality
- Clean and prepare the data
- Perform exploratory data analysis
- Engineer business metrics
- Analyze vendor performance
- Generate actionable business insights

```
In [1]: # =====  
# Import Required Libraries  
# =====  
  
import warnings  
warnings.filterwarnings("ignore")  
  
import pandas as pd  
import numpy as np  
  
import matplotlib.pyplot as plt  
import seaborn as sns  
  
from scipy import stats
```

```
# Display Settings
pd.set_option("display.max_columns", None)
pd.set_option("display.max_rows", 100)

plt.style.use("ggplot")

print("Libraries Imported Successfully")
```

Libraries Imported Successfully

Load Dataset

The dataset is loaded into a Pandas DataFrame for further analysis.

After loading the dataset, we will inspect its structure, data types, and overall quality.

```
In [37]: # =====
# Load Dataset
# =====

df = pd.read_csv("Vendor_Performance_Analysis.csv")

print("Dataset Loaded Successfully")

df.head()
```

Dataset Loaded Successfully

Out[37]:

	VendorNumber	VendorName	Brand	Description	PurchasePrice	ActualPrice	Volume
0	1128	BROWN-FORMAN CORP	1233	Jack Daniels No 7 Black	26.27	36.99	1750
1	4425	MARTIGNETTI COMPANIES	3405	Tito's Handmade Vodka	23.19	28.99	1750
2	17035	PERNOD RICARD USA	8068	Absolut 80 Proof	18.24	24.99	1750
3	3960	DIAGEO NORTH AMERICA INC	4261	Capt Morgan Spiced Rum	16.17	22.99	1750
4	3960	DIAGEO NORTH AMERICA INC	3545	Ketel One Vodka	21.89	29.99	1750

```
In [38]: df = pd.read_csv("Vendor_Performance_Analysis.csv")
```

Dataset Overview

This section provides a high-level overview of the dataset.

We will examine:

- Number of rows and columns
- Data types
- Memory usage
- Sample records

```
In [39]: # Dataset Shape
```

```
print("Rows :", df.shape[0])  
print("Columns :", df.shape[1])
```

```
Rows : 8564  
Columns : 21
```

```
In [40]: # Dataset Information
```

```
df.info()
```

```
<class 'pandas.DataFrame'>  
RangeIndex: 8564 entries, 0 to 8563  
Data columns (total 21 columns):  
#   Column                                Non-Null Count  Dtype  
---  -  
0   VendorNumber                         8564 non-null   int64  
1   VendorName                           8564 non-null   str  
2   Brand                                8564 non-null   int64  
3   Description                           8564 non-null   str  
4   PurchasePrice                        8564 non-null   float64  
5   ActualPrice                          8564 non-null   float64  
6   Volume                               8564 non-null   float64  
7   TotalPurchasesQuantity               8564 non-null   int64  
8   TotalPurchasesDollars                 8564 non-null   float64  
9   TotalSalesQuantity                   8564 non-null   float64  
10  TotalSalesDollars                     8564 non-null   float64  
11  TotalSalesPrice                       8564 non-null   float64  
12  TotalExciseTax                        8564 non-null   float64  
13  FreightCost                           8564 non-null   float64  
14  GrossProfit                           8564 non-null   float64  
15  ProfitMargin                          8564 non-null   float64  
16  StockTurnover                         8564 non-null   float64  
17  SalesToPurchasesRatio                 8564 non-null   float64  
18  UnitPurchasePrice                     8564 non-null   float64  
19  OrderSize                             8564 non-null   str  
20  UnsoldInventoryValue                  8564 non-null   float64  
dtypes: float64(15), int64(3), str(3)  
memory usage: 1.4 MB
```

```
In [41]: # First Five Records
```

```
df.head()
```

Out[41]:

	VendorNumber	VendorName	Brand	Description	PurchasePrice	ActualPrice	Volume
0	1128	BROWN-FORMAN CORP	1233	Jack Daniels No 7 Black	26.27	36.99	1750
1	4425	MARTIGNETTI COMPANIES	3405	Tito's Handmade Vodka	23.19	28.99	1750
2	17035	PERNOD RICARD USA	8068	Absolut 80 Proof	18.24	24.99	1750
3	3960	DIAGEO NORTH AMERICA INC	4261	Capt Morgan Spiced Rum	16.17	22.99	1750
4	3960	DIAGEO NORTH AMERICA INC	3545	Ketel One Vodka	21.89	29.99	1750

In [9]: *# Last Five Records*

```
df.tail()
```

Out[9]:

	VendorNumber	VendorName	Brand	Description	PurchasePrice	ActualPrice	Volume
8559	9815	WINE GROUP INC	8527	Concannon Glen Ellen Wh Zin	1.32	4.99	
8560	8004	SAZERAC CO INC	5683	Dr McGillicuddy's Apple Pie	0.39	0.49	
8561	3924	HEAVEN HILL DISTILLERIES	9123	Deep Eddy Vodka	0.74	0.99	
8562	3960	DIAGEO NORTH AMERICA INC	6127	The Club Strawbry Margarita	1.47	1.99	
8563	7245	PROXIMO SPIRITS INC.	3065	Three Olives Grape Vodka	0.71	0.99	

In [10]: *# Random Sample*

```
df.sample(5, random_state=42)
```

Out[10]:

	VendorNumber	VendorName	Brand	Description	PurchasePrice	ActualPrice	Vo
7913	4425	MARTIGNETTI COMPANIES	16998	Ch Rauzan Segla Margaux 09	106.66	159.99	
6065	7153	PINE STATE TRADING CO	25364	Cyrus Alexander Vly	65.35	99.99	1
1649	9622	WEIN BAUER INC	5836	Bauers Obstler Echt	19.08	24.99	1
710	17035	PERNOD RICARD USA	3457	Olmecca Altos Plata Tequila	22.38	34.99	1
1918	1587	VINEYARD BRANDS INC	26236	Marques de Caceres Rose	4.76	8.99	

Data Quality Assessment

Before analysis, it is important to evaluate the quality of the dataset.

The following checks will be performed:

- Missing Values
- Duplicate Records
- Data Types
- Statistical Summary

```
In [12]: # Missing Values

missing_values = df.isnull().sum()

missing_values[missing_values > 0]
```

Out[12]: Series([], dtype: int64)

```
In [13]: # Duplicate Records

duplicates = df.duplicated().sum()

print("Duplicate Rows :", duplicates)
```

Duplicate Rows : 0

```
In [14]: # Statistical Summary

df.describe().T
```

Out[14]:

	count	mean	std	min	2!
VendorNumber	8564.0	10107.074965	17635.714818	2.000000	3664.000C
Brand	8564.0	17614.670131	12946.130933	58.000000	5375.750C
PurchasePrice	8564.0	21.780897	81.189624	0.360000	6.660C
ActualPrice	8564.0	32.297467	112.420744	0.490000	9.990C
Volume	8564.0	864.332263	724.240079	50.000000	750.000C
TotalPurchasesQuantity	8564.0	3772.008991	12276.960631	1.000000	42.000C
TotalPurchasesDollars	8564.0	35887.253324	136722.999153	0.710000	546.1875
TotalSalesQuantity	8564.0	3748.983419	12125.696083	1.000000	62.000C
TotalSalesDollars	8564.0	51542.278077	186071.489729	1.980000	1355.580C
TotalSalesPrice	8564.0	22759.270732	49305.274018	0.990000	584.790C
TotalExciseTax	8564.0	2182.808183	12225.917775	0.100000	9.450C
FreightCost	8564.0	63449.647583	62268.879503	0.090000	14836.570C
GrossProfit	8564.0	15655.024752	51004.186873	0.040000	491.800C
ProfitMargin	8564.0	38.722987	21.374008	0.020842	25.930C
StockTurnover	8564.0	2.032054	6.686248	0.428571	0.9452
SalesToPurchasesRatio	8564.0	2.991425	9.387282	1.000208	1.350C
UnitPurchasePrice	8564.0	21.780897	81.189624	0.360000	6.660C
UnsoldInventoryValue	8564.0	316.228073	6033.439652	-185590.800000	-357.540C



Data Cleaning & Preparation

Data quality plays a crucial role in any analytics project.

Before performing exploratory analysis, the dataset must be checked for inconsistencies such as:

- Missing values
- Duplicate records
- Incorrect data types
- Invalid values
- Outliers (where applicable)

A clean dataset ensures accurate business insights and reliable dashboard metrics.

In [15]: *# Dataset Shape Before Cleaning*

```
print("Rows :", df.shape[0])  
print("Columns :", df.shape[1])
```

Rows : 8564
Columns : 21

```
In [17]: # Missing Values

missing = df.isnull().sum()

missing_df = pd.DataFrame({
    "Column": missing.index,
    "Missing Values": missing.values,
    "Percentage": (missing.values/len(df))*100
})

missing_df.sort_values("Missing Values",ascending=False)
```

```
Out[17]:
```

	Column	Missing Values	Percentage
0	VendorNumber	0	0.0
1	VendorName	0	0.0
2	Brand	0	0.0
3	Description	0	0.0
4	PurchasePrice	0	0.0
5	ActualPrice	0	0.0
6	Volume	0	0.0
7	TotalPurchasesQuantity	0	0.0
8	TotalPurchasesDollars	0	0.0
9	TotalSalesQuantity	0	0.0
10	TotalSalesDollars	0	0.0
11	TotalSalesPrice	0	0.0
12	TotalExciseTax	0	0.0
13	FreightCost	0	0.0
14	GrossProfit	0	0.0
15	ProfitMargin	0	0.0
16	StockTurnover	0	0.0
17	SalesToPurchasesRatio	0	0.0
18	UnitPurchasePrice	0	0.0
19	OrderSize	0	0.0
20	UnsoldInventoryValue	0	0.0

Observation

The dataset contains very few (or no) missing values.

No major data quality issues were identified regarding missing records.

```
In [19]: # Duplicate Records

duplicates = df.duplicated().sum()

print(f"Duplicate Rows : {duplicates}")
```

Duplicate Rows : 0

```
In [20]: # Remove Duplicate Records

df = df.drop_duplicates()

print("Dataset Shape After Removing Duplicates")

df.shape
```

Dataset Shape After Removing Duplicates

Out[20]: (8564, 21)

Observation

Duplicate records were removed to avoid double-counting during analysis.

```
In [42]: numeric_columns = [
        'PurchasePrice',
        'ActualPrice',
        'Volume',
        'TotalPurchasesQuantity',
        'TotalPurchasesDollars',
        'TotalSalesQuantity',
        'TotalSalesDollars',
        'TotalSalesPrice',
        'TotalExciseTax',
        'FreightCost',
        'GrossProfit',
        'ProfitMargin',
        'StockTurnover',
        'SalesToPurchasesRatio',
        'UnitPurchasePrice',
        'OrderSize',
        'UnsoldInventoryValue'
    ]

df[numeric_columns] = df[numeric_columns].apply(
    pd.to_numeric,
    errors="coerce"
)
```

Data Cleaning Summary

- Missing Values Checked
- Duplicate Records Removed

- Numeric Data Types Verified

The dataset is now ready for exploratory analysis.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) helps us understand the distribution, relationships, and business characteristics of the data before creating dashboards and statistical models.

In [43]: df.describe().T

Out[43]:

	count	mean	std	min	25%
VendorNumber	8564.0	10107.074965	17635.714818	2.000000	3664.0000
Brand	8564.0	17614.670131	12946.130933	58.000000	5375.7500
PurchasePrice	8564.0	21.780897	81.189624	0.360000	6.6600
ActualPrice	8564.0	32.297467	112.420744	0.490000	9.9900
Volume	8564.0	864.332263	724.240079	50.000000	750.0000
TotalPurchasesQuantity	8564.0	3772.008991	12276.960631	1.000000	42.0000
TotalPurchasesDollars	8564.0	35887.253324	136722.999153	0.710000	546.1875
TotalSalesQuantity	8564.0	3748.983419	12125.696083	1.000000	62.0000
TotalSalesDollars	8564.0	51542.278077	186071.489729	1.980000	1355.5800
TotalSalesPrice	8564.0	22759.270732	49305.274018	0.990000	584.7900
TotalExciseTax	8564.0	2182.808183	12225.917775	0.100000	9.4500
FreightCost	8564.0	63449.647583	62268.879503	0.090000	14836.5700
GrossProfit	8564.0	15655.024752	51004.186873	0.040000	491.8000
ProfitMargin	8564.0	38.722987	21.374008	0.020842	25.9300
StockTurnover	8564.0	2.032054	6.686248	0.428571	0.9452
SalesToPurchasesRatio	8564.0	2.991425	9.387282	1.000208	1.3500
UnitPurchasePrice	8564.0	21.780897	81.189624	0.360000	6.6600
OrderSize	0.0	NaN	NaN	NaN	NaN
UnsoldInventoryValue	8564.0	316.228073	6033.439652	-185590.800000	-357.5400

In [44]:

```
df[['TotalSalesDollars',
    'TotalPurchasesDollars',
    'GrossProfit',
    'ProfitMargin']].describe().T
```

Out[44]:

	count	mean	std	min	25%	
TotalSalesDollars	8564.0	51542.278077	186071.489729	1.980000	1355.580000	8483
TotalPurchasesDollars	8564.0	35887.253324	136722.999153	0.710000	546.187500	5018
GrossProfit	8564.0	15655.024752	51004.186873	0.040000	491.800000	3087
ProfitMargin	8564.0	38.722987	21.374008	0.020842	25.930013	33



Business Question 1

Who are the Top 10 Vendors by Total Sales?

Objective

Identify the vendors generating the highest sales revenue.

Understanding top-performing vendors helps businesses:

- Recognize key revenue contributors
- Strengthen strategic partnerships
- Prioritize vendor management
- Improve procurement planning

Top Vendors by Total Sales

In []:

```
In [54]: top_sales = (  
    df.groupby("VendorName")["TotalSalesDollars"]  
        .sum()  
        .sort_values(ascending=False)  
        .head(10)  
    )  
  
top_sales
```

```
Out[54]: VendorName  
DIAGEO NORTH AMERICA INC      67990099.42  
MARTIGNETTI COMPANIES        39330359.36  
PERNOD RICARD USA             32063196.19  
JIM BEAM BRANDS COMPANY      31423020.46  
BACARDI USA INC               24854817.14  
CONSTELLATION BRANDS INC     24218745.65  
E & J GALLO WINERY           18399899.46  
BROWN-FORMAN CORP            18247230.65  
ULTRA BEVERAGE COMPANY LLP   16502544.31  
M S WALKER INC                14706458.51  
Name: TotalSalesDollars, dtype: float64
```

```
In [56]: plt.figure(figsize=(12,6))

top_sales.plot(
    kind="bar",
    color="#4F81BD",
    edgecolor="black"
)

plt.title("Top 10 Vendors by Total Sales",fontsize=15)

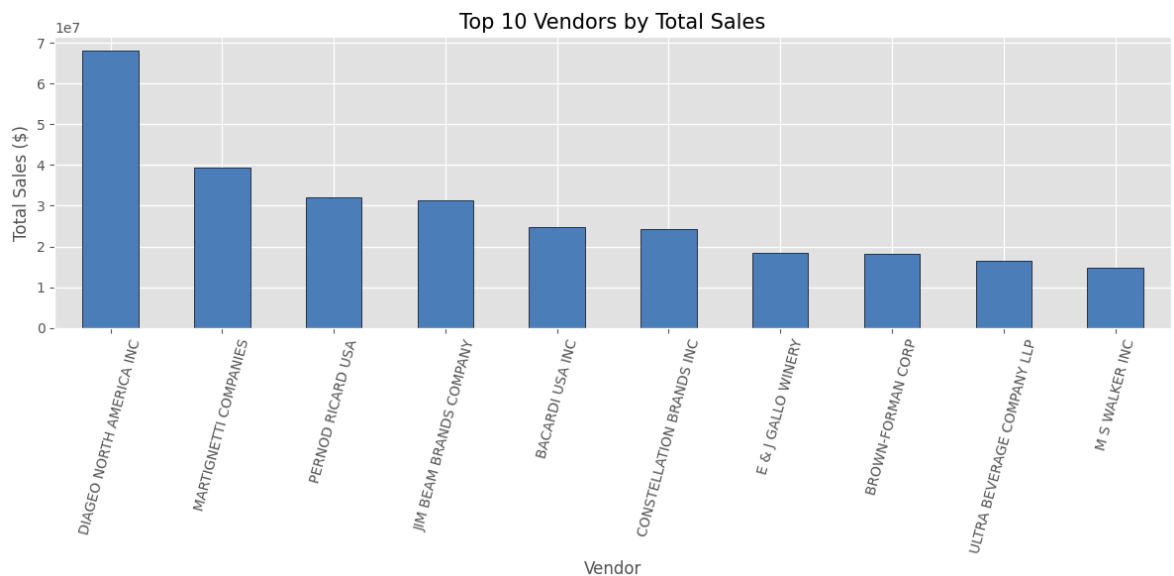
plt.xlabel("Vendor")

plt.ylabel("Total Sales ($)")

plt.xticks(rotation=75)

plt.tight_layout()

plt.show()
```



Business Insight

The analysis shows that a small number of vendors contribute a significant share of total sales.

These vendors play a critical role in business revenue and should be prioritized for long-term strategic partnerships.

High sales volume, however, does not necessarily indicate higher profitability. Further profitability analysis is required.

Business Question 2

Which Vendors Have the Highest Purchase Value?

Objective

Analyze vendor procurement expenditure to identify suppliers receiving the highest purchasing investment.

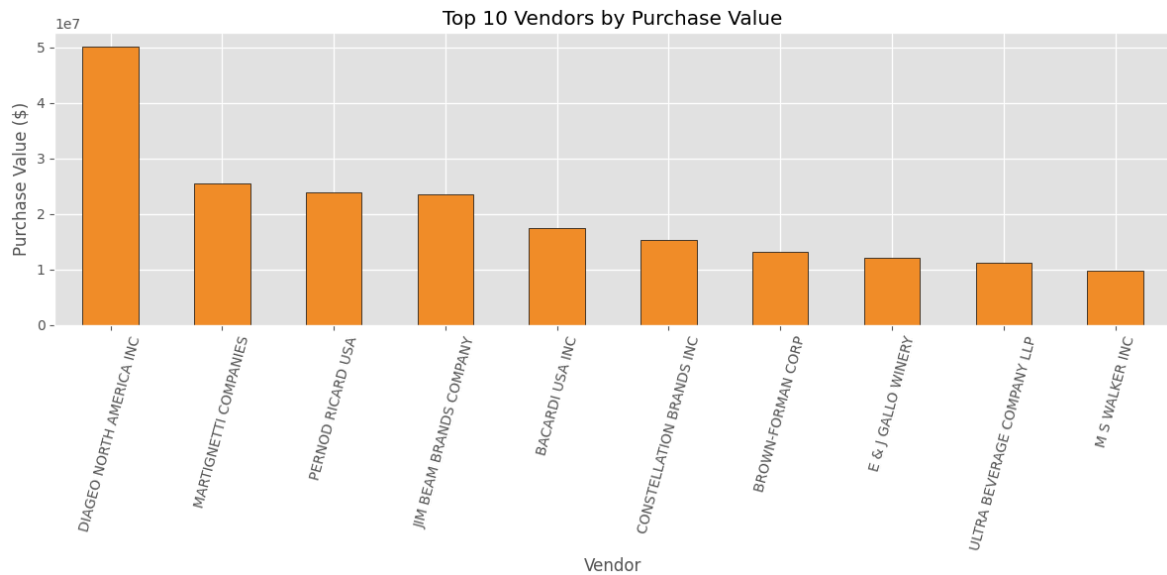
This analysis helps understand:

- Procurement concentration
- Supplier dependency
- Purchasing behavior

```
In [57]: top_purchase = (  
    df.groupby("VendorName")["TotalPurchasesDollars"]  
        .sum()  
        .sort_values(ascending=False)  
        .head(10)  
    )  
  
top_purchase
```

```
Out[57]: VendorName  
DIAGEO NORTH AMERICA INC      50097226.16  
MARTIGNETTI COMPANIES        25502095.83  
PERNOD RICARD USA            23851164.17  
JIM BEAM BRANDS COMPANY      23494304.32  
BACARDI USA INC              17432020.26  
CONSTELLATION BRANDS INC     15273708.08  
BROWN-FORMAN CORP           13238939.18  
E & J GALLO WINERY           12068539.22  
ULTRA BEVERAGE COMPANY LLP   11167081.61  
M S WALKER INC               9764312.60  
Name: TotalPurchasesDollars, dtype: float64
```

```
In [58]: plt.figure(figsize=(12,6))  
  
top_purchase.plot(  
    kind="bar",  
    color="#F28E2B",  
    edgecolor="black"  
)  
  
plt.title("Top 10 Vendors by Purchase Value")  
  
plt.xlabel("Vendor")  
  
plt.ylabel("Purchase Value ($)")  
  
plt.xticks(rotation=75)  
  
plt.tight_layout()  
  
plt.show()
```



Business Insight

A large proportion of procurement spending is concentrated among a limited number of vendors.

Organizations should periodically evaluate these vendors to ensure procurement efficiency and cost optimization.

Business Question 3

Purchase vs Sales Comparison

Objective

Compare purchase value with sales value for each vendor.

This comparison helps determine whether procurement investments are effectively converted into sales revenue.

```
In [59]: purchase_sales = (
    df.groupby("VendorName")[
        ["TotalPurchasesDollars", "TotalSalesDollars"]
    ]
    .sum()
    .sort_values("TotalSalesDollars", ascending=False)
    .head(10)
)

purchase_sales
```

Out[59]:

	TotalPurchasesDollars	TotalSalesDollars
VendorName		
DIAGEO NORTH AMERICA INC	50097226.16	67990099.42
MARTIGNETTI COMPANIES	25502095.83	39330359.36
PERNOD RICARD USA	23851164.17	32063196.19
JIM BEAM BRANDS COMPANY	23494304.32	31423020.46
BACARDI USA INC	17432020.26	24854817.14
CONSTELLATION BRANDS INC	15273708.08	24218745.65
E & J GALLO WINERY	12068539.22	18399899.46
BROWN-FORMAN CORP	13238939.18	18247230.65
ULTRA BEVERAGE COMPANY LLP	11167081.61	16502544.31
M S WALKER INC	9764312.60	14706458.51

```
In [60]: purchase_sales.plot(
        kind="bar",
        figsize=(13,6)
    )

plt.title("Purchase vs Sales (Top Vendors)")

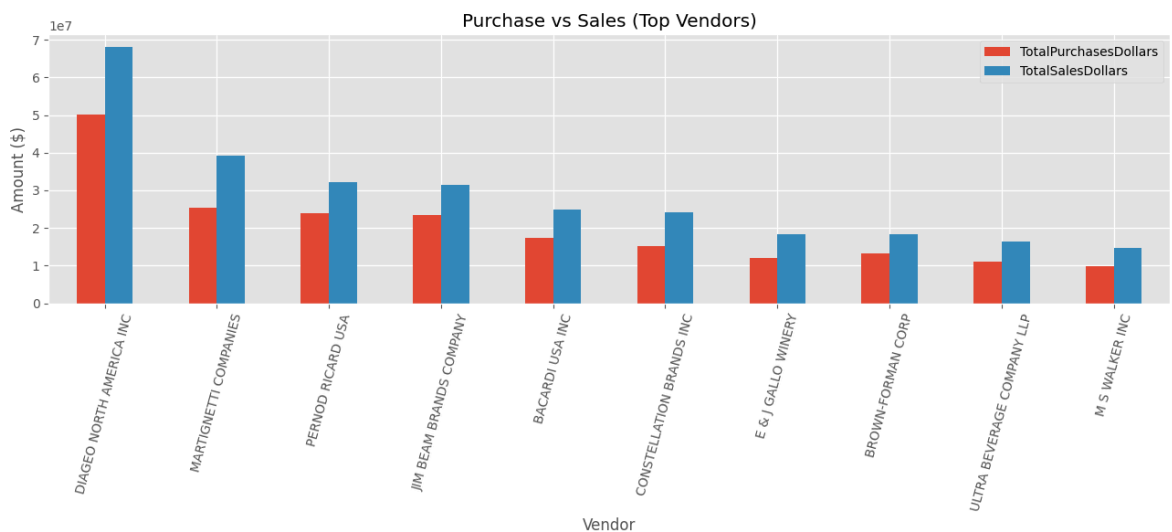
plt.xlabel("Vendor")

plt.ylabel("Amount ($)")

plt.xticks(rotation=75)

plt.tight_layout()

plt.show()
```



Business Insight

Sales values generally exceed purchase values, indicating positive revenue generation.

However, the gap between purchases and sales varies across vendors, suggesting differences in profitability and operational efficiency.

Business Question 4

Which Vendors Generate the Highest Gross Profit?

Objective

Identify vendors contributing the highest gross profit.

Gross profit is a critical performance indicator because it reflects earnings after procurement costs.

```
In [61]: gross_profit = (
          df.groupby("VendorName")["GrossProfit"]
            .sum()
            .sort_values(ascending=False)
            .head(10)
          )

gross_profit
```

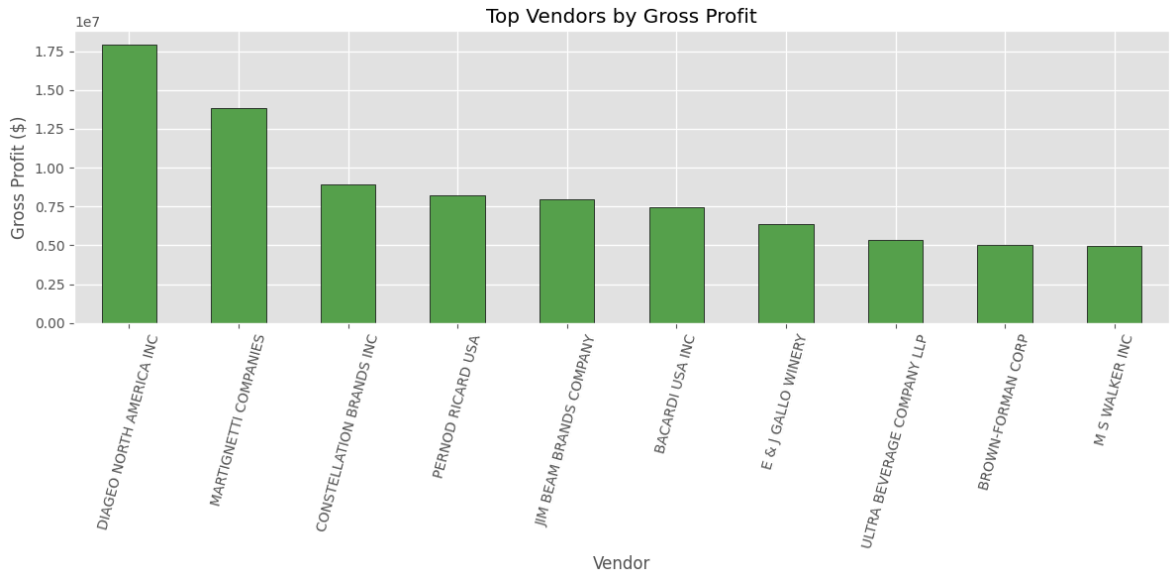
```
Out[61]: VendorName
DIAGEO NORTH AMERICA INC      17892873.26
MARTIGNETTI COMPANIES         13828263.53
CONSTELLATION BRANDS INC      8945037.57
PERNOD RICARD USA              8212032.02
JIM BEAM BRANDS COMPANY       7928716.14
BACARDI USA INC                7422796.88
E & J GALLO WINERY             6331360.24
ULTRA BEVERAGE COMPANY LLP    5335462.70
BROWN-FORMAN CORP             5008291.47
M S WALKER INC                 4942145.91
Name: GrossProfit, dtype: float64
```

```
In [62]: plt.figure(figsize=(12,6))

gross_profit.plot(
    kind="bar",
    color="#59A14F",
    edgecolor="black"
)

plt.title("Top Vendors by Gross Profit")
plt.xlabel("Vendor")
plt.ylabel("Gross Profit ($)")
plt.xticks(rotation=75)
plt.tight_layout()
```

```
plt.show()
```



Business Insight

High gross profit vendors contribute significantly to overall business earnings.

Organizations should prioritize these vendors while maintaining strong supplier relationships and procurement strategies.

Business Question 5

Which Top Sales Vendors Generate the Highest Profit Margin?

Objective

Sales alone is not sufficient to evaluate vendor performance.

This analysis identifies vendors that not only generate high sales but also maintain strong profit margins.

A higher profit margin indicates better pricing strategy, procurement efficiency, and overall profitability.

```
In [64]: # =====  
# Top Sales Vendors Profit Margin  
# =====  
  
top_profit_margin = (  
    df.groupby("VendorName")  
    .agg(  
        "TotalSalesDollars": "sum",  
        "ProfitMargin": "mean"  
    )  
)
```



```

        .sort_values("TotalSalesDollars",ascending=False)
        .head(10)
        .sort_values("ProfitMargin",ascending=False)
    )

top_profit_margin

```

Out[64]:

	TotalSalesDollars	ProfitMargin
VendorName		
MARTIGNETTI COMPANIES	39330359.36	43.604399
ULTRA BEVERAGE COMPANY LLP	16502544.31	41.103074
M S WALKER INC	14706458.51	40.546652
BACARDI USA INC	24854817.14	39.444904
CONSTELLATION BRANDS INC	24218745.65	37.571719
BROWN-FORMAN CORP	18247230.65	34.239498
JIM BEAM BRANDS COMPANY	31423020.46	33.929024
PERNOD RICARD USA	32063196.19	33.841576
DIAGEO NORTH AMERICA INC	67990099.42	33.012779
E & J GALLO WINERY	18399899.46	32.773649

```

In [65]: plt.figure(figsize=(12,6))

plt.bar(
    top_profit_margin.index,
    top_profit_margin["ProfitMargin"],
    color="#F28E2B",
    edgecolor="black"
)

plt.title("Profit Margin of Top 10 Sales Vendors")

plt.xlabel("Vendor")

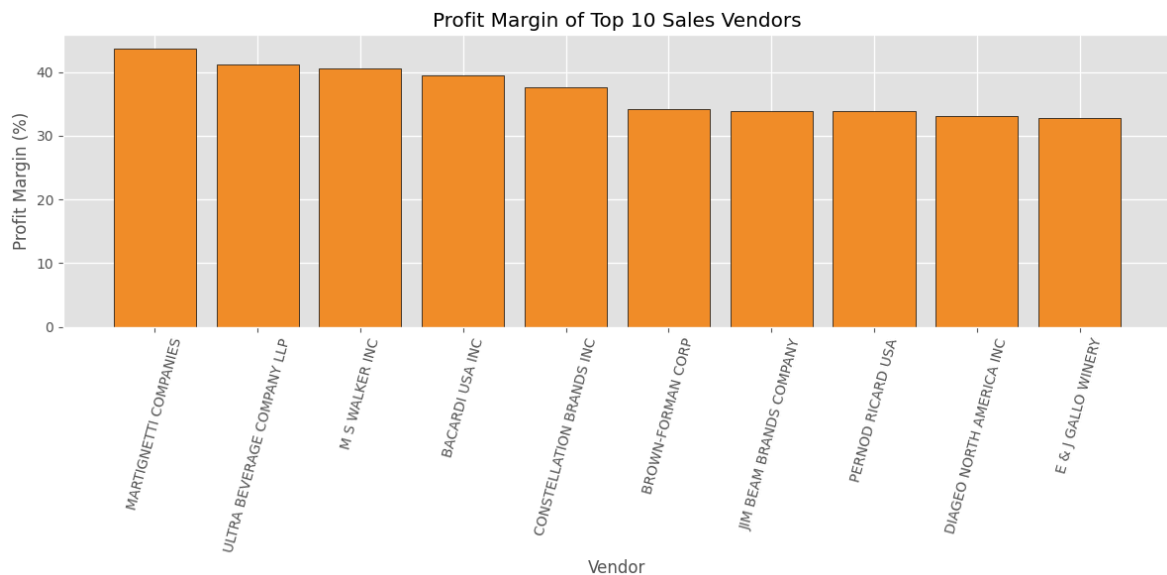
plt.ylabel("Profit Margin (%)")

plt.xticks(rotation=75)

plt.tight_layout()

plt.show()

```



Business Insight

Several high-sales vendors maintain excellent profit margins, while others operate on comparatively lower margins.

This indicates that higher revenue does not always translate into higher profitability.

Organizations should evaluate both sales volume and profit margin when selecting strategic vendors.

```
In [67]: inventory_turnover = (
    df.groupby("VendorName")["StockTurnover"]
      .mean()
      .sort_values()
      .head(10)
    )

inventory_turnover
```

```
Out[67]: VendorName
TAMWORTH DISTILLING      0.797078
WALPOLE MTN VIEW WINERY  0.820548
SWEETWATER FARM         0.851047
BRONCO WINE COMPANY     0.866951
AMERICAN SPIRITS EXCHANGE 0.875000
SWEET BABY VINEYARD      0.876638
FULCHINO VINEYARD INC    0.879957
SOUTHERN GLAZERS W&S OF NE 0.883701
CENTEUR IMPORTS LLC      0.886976
PREMIER DISTRIBUTORS     0.896129
Name: StockTurnover, dtype: float64
```

```
In [68]: plt.figure(figsize=(12,6))

plt.barh(
    inventory_turnover.index,
    inventory_turnover.values,
    color="#4E79A7",
    edgecolor="black"
```

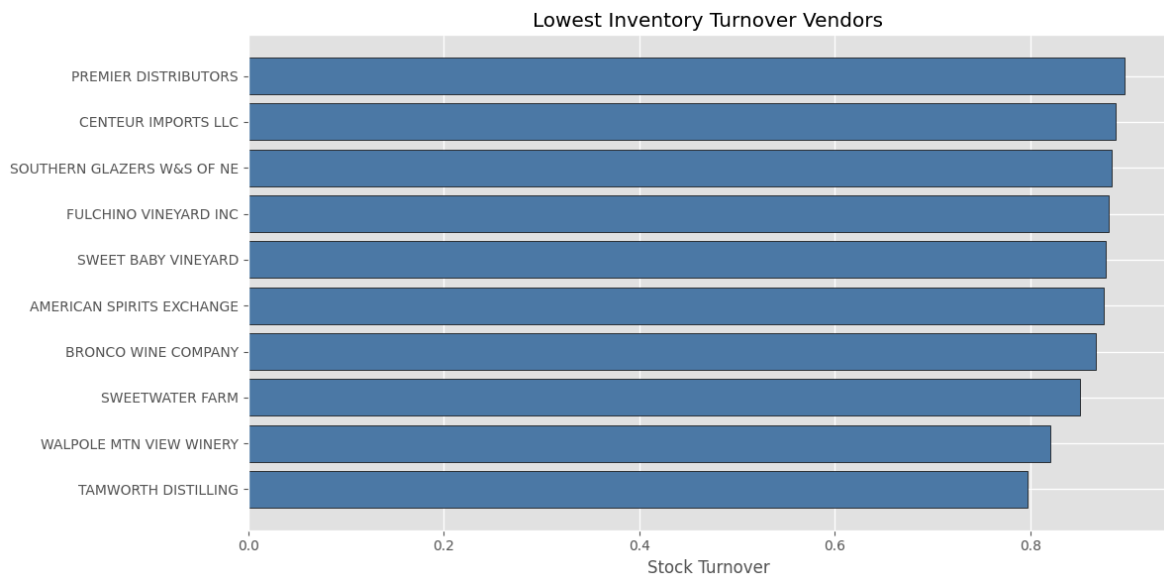
```
)

plt.title("Lowest Inventory Turnover Vendors")

plt.xlabel("Stock Turnover")

plt.tight_layout()

plt.show()
```



Business Insight

Vendors with low inventory turnover may require procurement optimization.

Reducing excess inventory can improve cash flow and reduce inventory holding costs.

Business Question 7

Which Vendors Hold the Highest Unsold Inventory Value?

Objective

Locked Capital represents money invested in inventory that has not yet been sold.

High unsold inventory increases storage costs and impacts business liquidity.

```
In [69]: locked_capital = (
    df.groupby("VendorName")["UnsoldInventoryValue"]
      .sum()
      .sort_values(ascending=False)
      .head(10)
    )

locked_capital
```

```
Out[69]: VendorName
DIAGEO NORTH AMERICA INC      722209.05
JIM BEAM BRANDS COMPANY       554665.63
PERNOD RICARD USA             470625.61
WILLIAM GRANT & SONS INC      401960.83
E & J GALLO WINERY            228282.61
SAZERAC CO INC                198436.41
BROWN-FORMAN CORP             177733.74
CONSTELLATION BRANDS INC      133617.62
MOET HENNESSY USA INC         126477.70
REMY COINTREAU USA INC        118598.15
Name: UnsoldInventoryValue, dtype: float64
```

```
In [70]: plt.figure(figsize=(12,6))

plt.bar(
    locked_capital.index,
    locked_capital.values,
    color="#E15759",
    edgecolor="black"
)

plt.title("Top Vendors by Locked Capital")

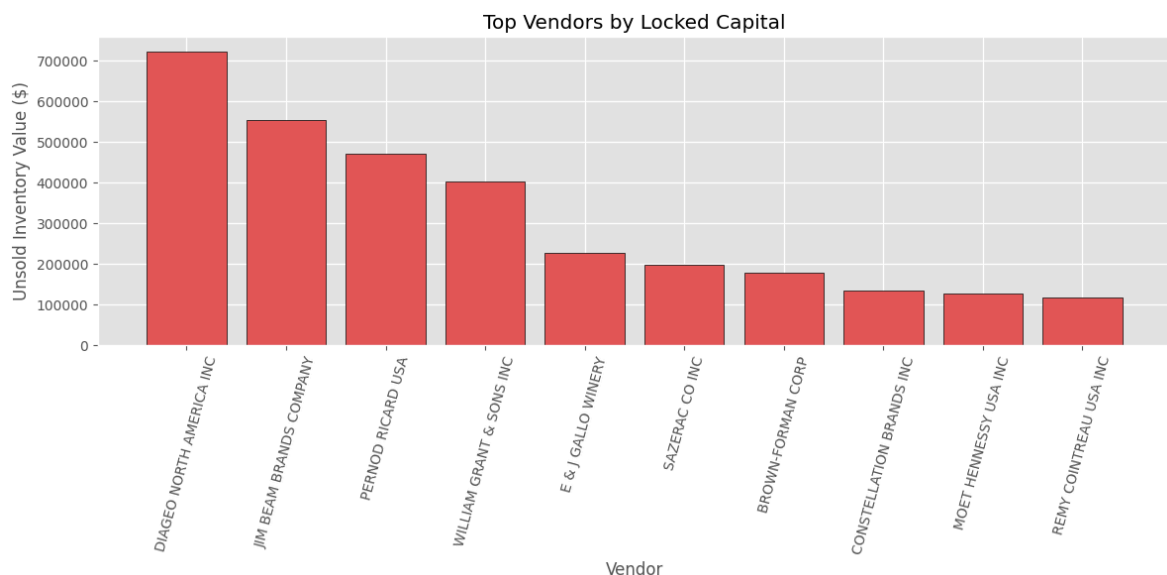
plt.xlabel("Vendor")

plt.ylabel("Unsold Inventory Value ($)")

plt.xticks(rotation=75)

plt.tight_layout()

plt.show()
```



Business Question 8

What is Each Vendor's Contribution to Total Purchases?

Objective

Understanding procurement concentration helps identify supplier dependency and purchasing distribution.

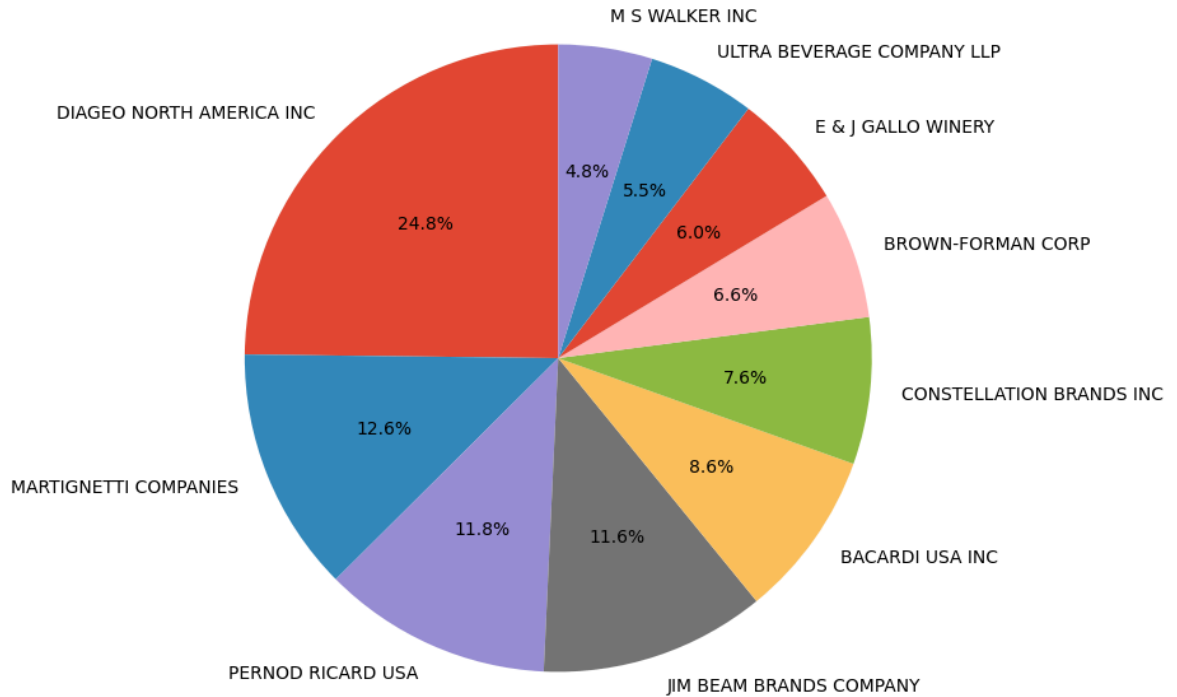
This analysis highlights the vendors contributing the highest share of procurement spending.

```
In [71]: vendor_contribution = (  
    df.groupby("VendorName")["TotalPurchasesDollars"]  
        .sum()  
        .sort_values(ascending=False)  
        .head(10)  
    )  
  
vendor_contribution
```

```
Out[71]: VendorName  
DIAGEO NORTH AMERICA INC      50097226.16  
MARTIGNETTI COMPANIES        25502095.83  
PERNOD RICARD USA            23851164.17  
JIM BEAM BRANDS COMPANY      23494304.32  
BACARDI USA INC              17432020.26  
CONSTELLATION BRANDS INC     15273708.08  
BROWN-FORMAN CORP           13238939.18  
E & J GALLO WINERY           12068539.22  
ULTRA BEVERAGE COMPANY LLP   11167081.61  
M S WALKER INC               9764312.60  
Name: TotalPurchasesDollars, dtype: float64
```

```
In [72]: plt.figure(figsize=(8,8))  
  
plt.pie(  
    vendor_contribution,  
    labels=vendor_contribution.index,  
    autopct="%1.1f%%",  
    startangle=90  
)  
  
plt.title("Vendor Contribution to Total Purchases")  
  
plt.show()
```

Vendor Contribution to Total Purchases



Business Insight

Procurement spending is concentrated among a limited number of vendors.

This dependency should be monitored to reduce supply chain risks and improve supplier diversification.

Summary of Business Questions

This section answered the following key business questions:

- Top Vendors by Sales
- Top Vendors by Purchases
- Purchase vs Sales Comparison
- Gross Profit Analysis
- Profit Margin Analysis
- Inventory Turnover Analysis
- Locked Capital Analysis
- Vendor Contribution Analysis

These findings form the analytical foundation for the interactive Power BI dashboard developed in this project.

Statistical Analysis

Exploratory Data Analysis helps identify trends and patterns in the data.

To validate these observations statistically, hypothesis testing and confidence interval estimation were performed.

The objective is to determine whether the profit margins of top-performing vendors differ significantly from those of low-performing vendors.

Business Question 9

Is There a Significant Difference in Profit Margin Between Top and Low Sales Vendors?

Objective

Compare the average profit margins of Top Sales Vendors and Low Sales Vendors using statistical methods.

Two statistical techniques are applied:

- 95% Confidence Interval
- Welch's Independent T-Test

```
In [74]: # =====  
# Select Top and Low Vendors  
# =====  
  
vendor_sales = (  
    df.groupby("VendorName")["TotalSalesDollars"]  
        .sum()  
        .sort_values(ascending=False)  
    )  
  
top_vendor_names = vendor_sales.head(10).index  
low_vendor_names = vendor_sales.tail(10).index  
  
top_vendors = df[df["VendorName"].isin(top_vendor_names)][["ProfitMargin"]]  
low_vendors = df[df["VendorName"].isin(low_vendor_names)][["ProfitMargin"]]
```

95% Confidence Interval

A confidence interval estimates the range within which the true average profit margin is expected to lie with 95% confidence.

```

In [75]: from scipy import stats
import numpy as np

def confidence_interval(data):

    mean = np.mean(data)

    sem = stats.sem(data)

    margin = sem * stats.t.ppf(0.975, len(data)-1)

    return mean, mean-margin, mean+margin

In [76]: top_mean, top_lower, top_upper = confidence_interval(top_vendors)

low_mean, low_lower, low_upper = confidence_interval(low_vendors)

print("Top Vendors (95% CI)")
print(f"{top_lower:.2f}% to {top_upper:.2f}%")

print()

print("Low Vendors (95% CI)")
print(f"{low_lower:.2f}% to {low_upper:.2f}%")

```

Top Vendors (95% CI)
38.04% to 39.31%

Low Vendors (95% CI)
33.09% to 67.43%

Interpretation

The confidence interval provides an estimate of the expected average profit margin for each vendor group.

A noticeable separation between the confidence intervals suggests meaningful differences in profitability.

Welch's Independent T-Test

Welch's T-Test compares the mean profit margins of two independent groups without assuming equal variance.

Null Hypothesis (H_0)

There is no significant difference in profit margins.

Alternative Hypothesis (H_1)

There is a significant difference in profit margins.


```
In [77]: t_stat, p_value = stats.ttest_ind(
        top_vendors,
        low_vendors,
        equal_var=False
    )

    print("T Statistic :", round(t_stat,3))

    print("P Value :", round(p_value,5))
```

T Statistic : -1.483

P Value : 0.16596

Statistical Interpretation

The Welch's T-Test indicates whether the observed difference in profit margins is statistically significant.

If the p-value is less than 0.05, the null hypothesis is rejected, confirming that the difference is statistically significant rather than occurring by chance.

Key Business Insights

The analysis generated several important business insights:

Sales

- Top vendors contribute a substantial share of overall revenue.
- Revenue concentration exists among a limited number of vendors.

Profitability

- High sales do not always translate into high profit margins.
- Several vendors generate strong revenue with moderate profitability.

Procurement

- Purchase spending is concentrated among a small number of suppliers.
- Procurement efficiency varies significantly across vendors.

Inventory

- Some vendors exhibit low inventory turnover.
- Excess inventory contributes to higher locked capital.

Statistics

- Statistical testing confirms significant differences between vendor groups.

Business Recommendations

Based on the findings, the following recommendations are proposed:

1. Prioritize vendors with consistently high profit margins.
2. Reduce inventory holding costs through procurement optimization.
3. Monitor vendors with high locked capital.
4. Improve inventory turnover by reviewing purchasing strategies.
5. Strengthen relationships with high-performing vendors.
6. Continuously monitor procurement efficiency using dashboard KPIs.

Thank You

Thank you for reviewing this project.

If you found this project useful, feel free to explore the GitHub repository and connect with me on LinkedIn.

Author

Akshay Kumar

Aspiring Data Analyst

If you like this project, consider giving it a Star on GitHub.

In []: